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GRADE: Gap Year

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RESEARCH TOPIC: HEPA Filtration for $PM_{2.5}$ Reduction in Dhaka

Keywords—*PM2.5, HEPA filtration, indoor air quality, GIS, hotspot analysis, Dhaka*

Title: Evaluating HEPA for Reducing Indoor $PM_{2.5}$ Exposure in Dhaka: A Low Cost Approach to Urban Air Pollution Mitigation

Abstract

Fine particulate matter (PM2.5) is a leading global health risk: in 2021 it contributed to about 7.83 million deaths worldwide, and South Asia carried one of the heaviest burdens [1]. In a study of 17 Dhaka homes, dry-season outdoor PM2.5 averaged 153 µg/m³ and indoor PM2.5 averaged 146 µg/m³ [2]. Portable air cleaners with high-efficiency particulate air (HEPA) filters lowered indoor PM2.5 by a mean of 49% across field studies worldwide [3], but the only Dhaka evidence found for this proposal is a small household pilot [4], and none of the sources reviewed tested where limited devices should be placed first. This proposal describes a before-and-after field study of HEPA air cleaners in 200 homes, offices, and classrooms in Dhaka city. PM2.5 will be logged with locally calibrated low-cost sensors before and during HEPA use, and a geographic information system (GIS) hotspot analysis will guide which sites are prioritized. The study will estimate the change in indoor PM2.5, test whether HEPA filtration removes PM2.5 in a statistically meaningful way, and compare GIS-prioritized sites with other sites in effectiveness and efficiency. The results are expected to give schools, employers, and city authorities an evidence-based way to allocate air-cleaning resources.

INTRODUCTION

A. Background

Exposure to PM2.5 contributed to about 7.83 million deaths and 231.5 million disability-adjusted life years worldwide in 2021, and South Asia and sub-Saharan Africa showed the highest disease burdens [1]. Bangladesh lies inside this high-burden region, and Dhaka has been described as one of the cities with the worst air pollution in the world [4]. The World Health Organization (WHO) annual PM2.5 guideline is 5 µg/m³, while Bangladesh's national annual and 24-h standards are 35 and 65 µg/m³; in 2022 the annual average at every ground station in a national analysis exceeded the national annual standard, and PM2.5 hot spots clustered around the Dhaka division [5]. In a study of 17 Dhaka homes, outdoor PM2.5 averaged 153 µg/m³ in the dry season (December–February) and 50 µg/m³ in the wet season [2].

B. Why Indoor Air Matters in Dhaka

In much of the world, people spend about 65% of their time indoors at home [6]. In Dhaka, the indoor-to-outdoor (I/O) PM2.5 ratio was 0.97 in the dry season and 1.05 in the wet season, and roughly 80–85% of indoor PM2.5 was estimated to come from outdoor infiltration [2]. None of the monitored homes had an air purifier, and indoor monitoring data remain scarce in Bangladesh [2]. Indoor spaces in Dhaka are therefore not a refuge from outdoor pollution, which makes indoor air cleaning a practical measure to test.

LITERATURE REVIEW

A. HEPA FILtration and Indoor $PM_{2.5}$

A HEPA filter captures at least 99.97% of 0.3 µm particles, the most penetrating size, and captures larger and smaller particles even more efficiently [7]. Field results vary widely. A systematic review of portable air purifiers in homes, schools, and offices reported PM2.5 reductions of 22.6% to 92.0% compared with controls [8]. A meta-analysis of 148 field studies found HEPA to be the most commonly studied technology and a mean PM2.5 reduction of 49% (SD 20%) across the 83 studies that measured it, and it warned that occupants often run devices briefly and on low settings because of noise, drafts, and electricity costs [3]. In real bedrooms, purifiers cut PM2.5 by 45% on average after 90 min whether windows were open or closed [6], and in school gyms they produced significant reductions that were larger with windows closed [9].

B. Evidence from Dhaka

Local evidence on purifiers is thin. A registered-report pilot in 41 households in Mirpur, Dhaka, found that offering an air purifier lowered indoor PM2.5 by 74 µg/m³ (27%) against a control mean of 279 µg/m³, so levels stayed far above WHO guidance, and purifier use was uneven across households [4]. Because that estimate reflects an offer of a purifier rather than measured use, it is an intent-to-treat effect. Outside the home, the Dhaka sources reviewed here did not include an evaluation of purifiers in offices or classrooms.

C. GIS, Hotspots, and Low-Cost Sensors

Hotspot analysis is used to decide where to act first. In India, Getis-Ord Gi* analysis of 1 km gridded PM2.5 identified consistent and emerging hotspots, many outside major cities, and the authors concluded that such maps can guide targeted action and resource allocation [11]. In Bangladesh, a national study using ground-station and satellite-derived data found positive spatial autocorrelation and suggested using hot spot maps to direct pollution-reduction efforts [5]. Within Dhaka, outdoor PM2.5 at homes was associated with restaurant density (R = 0.47), major-road density (R = 0.46), and population density (R = 0.28) within 1 km, which supports building a GIS layer from these predictors [2]. Low-cost sensors make dense monitoring affordable but need correction: uncorrected PurpleAir readings overestimated PM2.5 by about 40% in a large U.S. evaluation, and a correction with a humidity term cut the root mean square error from 8 to 3 µg/m³ [12]. In Dhaka, local co-location with a reference monitor gave a normalized error of 14% in the dry season and 17% in the wet season [2].

RESEARCH GAP AND AIM

The sources reviewed agree on three points: indoor air in Dhaka is about as polluted as outdoor air [2]; HEPA purifiers reduce PM2.5 but with wide variation [3], [8]; and hotspot mapping is an accepted way to prioritize action [5], [11]. They leave a gap. Within the sources reviewed for this proposal, none evaluated whether GIS-guided placement of HEPA units improves effectiveness and efficiency, and the only Dhaka purifier evidence found is a small household pilot [4]. This proposal addresses that gap. Its aim is to measure the change in indoor PM2.5 after HEPA installation in 200 homes, offices, and classrooms in Dhaka city, and to test whether GIS-guided placement is more effective and efficient.

RESEARCH QUESTIONS

RQ1. What are the indoor PM2.5 concentrations before and after HEPA air cleaners are installed and used in homes, offices, and classrooms in Dhaka?

RQ2. Does the use of HEPA air cleaners remove PM2.5 from indoor air to a statistically significant degree?

RQ3. Does GIS help place HEPA air cleaners where they are most needed, and is this placement effective and efficient?

RESEARCH METHODOLOGY

A. Study Design and Sites

The study uses a before-and-after (pre–post) field design in Dhaka city. Each of the 200 sites is its own control: PM2.5 is logged for a baseline period without an air cleaner (proposed: 7 days), then for an intervention period with one HEPA unit running in the main occupied room (proposed: 7 days). The proposed allocation is 80 homes, 60 offices, and 60 classrooms. One identical HEPA model at a fixed fan setting, sized to room volume, is used throughout, and sites that already own an air cleaner are excluded. Because the dry season has the highest PM2.5 [2], [5], fieldwork is planned for December–February, in five batches of 40 sites so that the same 40 sensor kits and 40 HEPA units can be rotated.

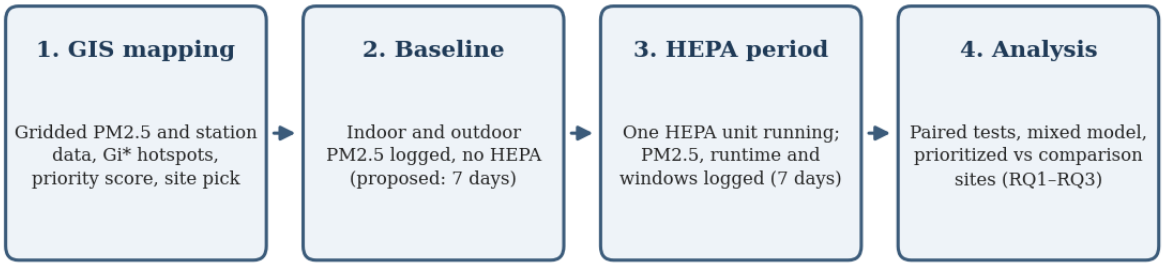


Fig. 1. Proposed study workflow.

B. GIS-Based Site Prioritization

GIS work is done before installation. First, candidate sites are geocoded and combined with gridded PM2.5 estimates and ground-station data, as used in the national Bangladesh analysis [5], together with layers for roads, restaurants, and population density [2]. Second, following the approach in [11], Getis-Ord Gi* hotspot analysis with neighbor-based spatial weights and Monte Carlo significance testing classifies each grid cell as a hotspot, coldspot, or not significant, with maps produced in QGIS. Third, each candidate site receives a priority score from its hotspot class and the number of occupants it serves. Within each setting, half of the sites are drawn from high-priority hotspot zones (GIS-prioritized group) and half from the remaining zones (comparison group), so the value of GIS targeting can be tested directly. Outdoor sensors placed during the campaign will show whether the mapped hotspots match measured outdoor PM2.5.

turnitin Page 9 of 12 - Integrity Submission Submission ID trn:oid:::3117:636712105

C. PM2.5 Measurement

Each site receives one low-cost optical PM2.5 sensor logging every 2 min (averaged to hourly and 24-h means), placed at breathing height away from the air cleaner outlet. Outdoor PM2.5 comes from a network of outdoor sensors and from the two continuous monitoring stations in Dhaka (Department of Environment and U.S. Embassy) [2]. Before fieldwork, all sensors are co-located with a reference-grade monitor to derive a local correction equation, as in [2] and [12]. Because sensors rotate across batches, week-to-week changes in city background PM2.5 are adjusted with the ratio method against a reference station, as in [2]. HEPA runtime and electricity use (plug-in energy logger), window opening, and major indoor sources such as cooking or smoking (short occupant diary) are also recorded, because user behavior strongly affects real-world performance [3].

D. Outcomes and Data Analysis

The primary outcome is mean indoor PM2.5 before and after HEPA use, expressed as absolute change (µg/m³) and percent reduction, (C_before − C_after) / C_before × 100. Indoor-to-outdoor ratios will be calculated so that changes in outdoor air between periods are not mistaken for filter effects [2]. **RQ1:** descriptive statistics and paired comparisons (paired *t*-test, or Wilcoxon signed-rank test if data are not normal) will be reported by setting. **RQ2:** a linear mixed-effects model with a random intercept for each site will estimate the HEPA effect while adjusting for outdoor PM2.5, setting, window state, and device runtime, with significance set at $p < 0.05$; results will be reported for all sites (intent-to-treat) and for sites that ran the unit as instructed. **RQ3:** GIS-prioritized and comparison sites will be compared on effectiveness (percent and absolute reduction) and efficiency (PM2.5 exposure avoided per device, in µg/m³ × occupant-hours, and per kWh used). The share of site-days with a 24-h mean below Bangladesh's 65 µg/m³ standard [5] will be compared before and after HEPA use, and the WHO annual guideline of 5 µg/m³ [5] will serve as a reference level, interpreted cautiously because monitoring is short-term.

TABLE I. ALIGNMENT OF RESEARCH QUESTIONS, DATA, AND ANALYSIS

RQ	Outcome	Main data	Analysis
RQ1	Indoor PM2.5 before vs after HEPA (µg/m³ and %), by setting	Indoor sensor, 24-h means	Descriptives; paired t-test or Wilcoxon test
RQ2	HEPA effect on indoor PM2.5, adjusted for outdoor PM2.5, setting, windows, runtime	Indoor and outdoor sensors, reference stations, energy logger, diary	Mixed-effects model; intent-to-treat and as-instructed
RQ3	Reduction and exposure avoided per device and per kWh, GIS-prioritized vs comparison sites	GIS layers, Gi* class, sensor and energy data	Group comparison of reductions and efficiency ratios

E. Ethics and Limitations

Participants will give informed consent, data will be anonymized, and approval will be sought from the responsible institutional ethics committee. The design has no sham-filter control, so weather-related confounding is possible; outdoor covariates, I/O ratios, and the reference-

station adjustment are used to limit it. Monitoring periods are short and limited to one dry season, one HEPA model is tested, a single sensor may not represent whole-room exposure, and sensor performance indoors was not separately calibrated in the Dhaka study used as a guide [2].

EXPECTED OUTCOMES AND SIGNIFICANCE

Significant PM2.5 reductions after HEPA use are expected, within the range reported in previous reviews [3], [8], although dry-season indoor levels in Dhaka may stay above guideline values even with a purifier, as the household pilot suggests [4]. If GIS targeting works as intended, hotspot sites should show larger absolute reductions and more exposure avoided per device than comparison sites. The study will produce (i) before-and-after PM2.5 evidence for three indoor settings in Dhaka, (ii) a reproducible GIS workflow for prioritizing sites, and (iii) practical guidance for schools, employers, and city authorities working with limited budgets.

PROPOSED TIMELINE

Period	Activity
Months 1–3	Ethics approval, site recruitment, GIS mapping and prioritization, sensor co-location and calibration
Months 4–6	Batch installation and monitoring in the dry season (baseline and HEPA periods)
Months 7–8	Data cleaning, statistical analysis, and reporting

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